1. **A task of analyzing database is given to the Database Administrator of the company having the following relational database and give an expression in relational algebra each of the following queries:**

   Employee (person-name, street, city)

   Works (person name, company name, salary)

   Company (Company name, city)

   Managers (person name, manager-name)

   a) Find the names of all employees who work for First Bank Corporation.

   b) Find the names and cities of residences of all employees who work for First Bank Corporation.

   c) Find the names of all the employees who do not work for First Bank Corporation.

   d) Find names of all employees who earn more than $10000 per annum.

2. **Consider a database with the following schema:**

Person (<u>name</u>, age, gender)	**name is a key**
Frequents (<u>name</u>, <u>pizzeria</u>)	**(name, pizzeria) is a key**
Eats (<u>name</u>, <u>pizza</u>)	**(name, pizza) is a key**
Serves (<u>pizzeria</u>, <u>pizza</u>, price)	**(pizzeria, pizza) is a key**

   Write relational algebra expressions for the following queries.

   a. Find all pizzerias frequented by at least one person under the age of 18.

   b. Find the names of all females who eat either mushroom or pepperoni pizza (or both).

   c. Find the names of all females who eat both mushroom and pepperoni pizza.

3. **A university maintains a database with the following relations:**

- **Student(SID, Name, Age, Major)**
- **Course(CID, Title, Credits, Department)**
- **Enrollment(SID, CID, Grade)**

   a) Retrieve the names of all students majoring in 'Computer Science'.

   b) List all students who have enrolled in 'Database Systems' (Assume CID for 'Database Systems' is known).

   c) Find students who have taken at least one course offered by the 'Mathematics' department.

   d) Get the names of students who have taken all courses offered by the 'Computer Science' department.

4. **An e-commerce website maintains the following relations:**

- **Customer(CID, Name, Age, City, Gender)**
- **Order(OID, CID, Date, TotalAmount)**
- **Product(PID, Name, Price, Category)**
- **OrderDetails(OID, PID, Quantity, Subtotal)**

   a) Find the names of customers who have placed orders worth more than $500.

b) Retrieve the details of all products purchased by a customer named 'Alice'.
c) Find customers who have never placed an order.
d) List all customers who have purchased at least one product from the 'Electronics' category.

5. **A library maintains records of books and their borrowings with these relations:**

- **Book(BID, Title, Author, Genre, Publisher, YearPublished)**
- **Member(MID, Name, Age, MembershipType, City)**
- **Borrowing(MID, BID, BorrowDate, ReturnDate)**

a) Find the names of members who have borrowed at least one book by 'J.K. Rowling'.
b) Retrieve the list of books that have never been borrowed.
c) Get the names of members who have borrowed all books of the genre 'Science Fiction'.
d) Find members who have borrowed books but never returned them.

6. **A hospital maintains patient and doctor information with these relations:**

- **Patient(PID, Name, Age, Gender, Disease, City)**
- **Doctor(DID, Name, Specialization, Experience, Department)**
- **Appointment(PID, DID, Date, Status)**

a) Retrieve the names of all patients who have consulted a 'Cardiologist'.
b) Find doctors who have treated more than 10 patients.
c) List all patients who have never booked an appointment.
d) Get the names of doctors who have treated patients with 'Diabetes'.